

YIXUAN ASHER ZHANG

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EDUCATION

University of Illinois Urbana-Champaign

M.S. in Computer Science

Champaign, IL

Aug 2026 – May 2028

University of Nottingham

B.Sc. (Hons) Computer Science, First Class · GPA 3.88/4.00

Nottingham, UK

Sep 2022 – Jun 2026

Honors: Outstanding Graduate (Top 5%); Dean's Scholarship (Top 10%); Zhejiang Provincial Government Scholarship (Top 5%); MCM Honorable Mention.

Coursework: Computer Vision (ranked 1st), Computer Security (ranked 1st), Operating Systems, Algorithms, Databases, Artificial Intelligence.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, Go, Swift

ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, computer vision, NLP, LoRA, RAG, ONNX, TensorRT, vLLM

Systems: FastAPI, Spring Boot, Docker, Kubernetes, MySQL, MLflow, GitHub Actions, AWS, GCP, Linux

SELECTED EXPERIENCE

Ningbo AST Machinery Technology

Ningbo, China

Software Engineer Intern | github.com/asher0913/multi-material-slicer

May 2026 – Aug 2026

- Shipped a cross-platform multi-material 3D-printing slicer in C++17, Qt, and OpenGL (STEP/STL assemblies, per-part materials, layer masks, G-code) with CMake builds and an Xvfb end-to-end self-test over the full import-to-G-code path.
- Designed a 51-table MySQL CRM (107 foreign keys, 70 constraints) and 10 Vue/FastAPI modules; built an LLM pipeline that produced 345 faculty profiles and 1,127 deduplicated grant records.

Zhiyang Innovation Technology

Shanghai, China

Machine Learning Engineer Intern

May 2025 – Aug 2025

- Cut Jetson Orin Nano inference latency from 220 ms to 95 ms (2.3×) and model size from 87 MB to 14 MB (6.2×) at under 0.5 mAP loss by exporting to ONNX and deploying INT8 TensorRT FP16 behind FastAPI and Docker.
- Raised mAP@0.5 from 0.74 to 0.79 and recall 6.2 points while cutting false positives 12% on a 7:1 imbalanced defect set; training pipeline for 120K+ images cut label noise 18% and data prep from 3 days to under 1, adding 5.6 mAP@0.5.

Beijing Corefire Technology

Beijing, China

Software Engineer Intern

May 2024 – Jul 2024

- Built merchant onboarding, query, and reporting services for UMFintech's acquiring platform in Spring Cloud and MySQL, covering hierarchical providers, POS/QR transactions, terminal lifecycle, and profit-sharing summaries.

LEADERSHIP

Students' Union, University of Nottingham

Nottingham, UK

Deputy Director

May 2023 – Apr 2024

- Organized a multilingual cultural season and film festival (400+ participants) and partnered with KPMG, NIO, Xiaomi, PwC, ByteDance, and Lufthansa on career sessions, each engaging 200+ students.

School of Computer Science, University of Nottingham

Nottingham, UK

Peer Mentor

Sep 2023 – Jan 2024

- Mentored 20+ first-years in weekly Java, recursion, data structures, complexity, and Git sessions; authored review sheets and earned helpful ratings from 90%+ of mentees.

SELECTED PROJECTS

DualPath-CEM | github.com/asher0913/DualPathCEM · PyTorch, privacy ML

Nov 2025 – Jun 2026

- Reached 81.96% mean top-1 on a 44,139-image FaceScrub split (526 identities, 5 seeds), 1.63 points above Noise_ARL+CEM, while raising decoder and GAN reconstruction MSE 75.8% and 74.9%.
- Designed dual-path split inference: a Gaussian-protected 16×16×16 SlotCEM tensor plus a 256-D MobileNetV3 token, fused by calibrated logits; SlotCEM projects 4,096-D smashed features into 8 slots of 64-D across a 64-entry per-class memory.

Manuscript in preparation: Y. Zhang and J. Ren, "DDP-CEM: Dual-Path Privacy Protection for Collaborative Face Inference," *Pattern Recognition*, 2026.

SkillRouter | github.com/asher0913/skill-router · Python, LLM agents

Sep 2026

- Cut per-query agent context from 14,108 to 2,052 tokens (−85.5%) on a 60-skill catalogue while holding Top-1 routing accuracy at 0.984 and Top-3 at 1.000; 26 tests in GitHub Actions.

InferenceLab | github.com/asher0913/inference-lab · Python, vLLM, MLOps

Jul 2026 – Sep 2026

- Built an async dynamic batcher and TTL/LRU semantic cache behind a swappable vLLM/SGLang/Ollama backend; an 80-request, 16-concurrency workload collapsed into 2 backend batches at 80% cache-hit rate and ~79 ms P95.

SecureRAG | github.com/asher0913/secure-rag · FastAPI, RAG, ACL

Feb 2025 – Apr 2025

- Implemented multi-tenant hybrid retrieval that applies tenant/ACL filters before ranking and repeats live permission checks before context assembly, failing closed on revocation; added Recall@K, MRR, and authorization-leakage tests.

CodeTaskForge | github.com/asher0913/code-task-forge · Python, FastAPI, agent eval

May 2025 – Jul 2025

- Built a sandboxed coding-agent eval pipeline (CLI + FastAPI) with shell-free command allowlists, timeouts, path-traversal defenses, and classified dependency/build/test/timeout/policy failures; Docker and CI regression coverage.

AgentGuard | github.com/asher0913/agent-guard · Python, FastAPI, AI security

Aug 2025 – Oct 2025

- Built an explainable pre-execution policy gateway (allow / approval-required / deny) for tool calls covering destructive shell, filesystem, exfiltration, and prompt injection; 100% precision and recall on the checked-in 10-case adversarial fixture.

Contactless Breathing Monitor | github.com/asher0913/contactless-breathing-monitor · Swift, TrueDepth

Aug 2024 – Oct 2024

- On-device iOS prototype syncing front-camera video and TrueDepth via AVCaptureMultiCamSession; Vision shoulder landmarks drive a chest ROI, then Gaussian filtering and a weighted moving average plot respiratory motion in SwiftUI.

- Fine-tuned ResNet-18 with mixed precision, cosine scheduling, EMA, and TTA to 89.8% test accuracy on 37 classes; Grad-CAM overlays, confusion matrices, and a 49.7% from-scratch residual CNN as the controlled baseline.